

MVGM — Methodology Version Governance Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Prediction Methodology Standard (PMES)

Operational Layer: Prediction Methodology Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Methodology Version Governance

Version: 1.0

Status: Canonical · Open Module

Effective Date: 6 September 2026

Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Methodology Version Governance

Canonical Definition

Methodology Version Governance Module (MVGM) defines the governance conditions under which a Prediction Methodology and its materially changed forms remain identifiable, distinguishable, versioned, historically preserved, traceable, and correctly associated with the predictions produced under them.

MVGM establishes whether a methodological change:

preserves the existing methodology version,

← requires a new version,

supersedes a prior version,

or:

creates a materially different Prediction Methodology.

The module prevents current methodology from silently replacing the methodology that actually governed an earlier prediction.

Methodology Update $\neq$ Historical Replacement

Operational Role

MVGM serves as the fifth and final internal governance space of the Prediction Methodology Standard (PMES) .

It receives:

Selected Predictive Method

Defined Methodological Structure

Documented Predictive Logic

Method Applicability State

and establishes the methodology version under which the Prediction Methodology may proceed into subsequent PGA™ governance.

The module asks:

Which exact version of the methodology governs this prediction, what changed from earlier versions, and can the methodology used at any historical point still be reconstructed?

The governed progression is:

Applicable Prediction Methodology → Version Identification → Change Assessment → Version Determination → Historical Preservation → Governed Prediction Methodology

The resulting governed methodology becomes an input to:

PBAS — Prediction Basis Standard.

Module Operational Space

MVGM governs:

methodology version identity; methodology version status; initial version formation; methodology change identification;

materiality of methodology change; minor methodological change; major methodological change; structural change; predictive logic change; applicability change; model replacement; model retraining; parameter change; threshold change; weighting change; aggregation change; human-role change; component replacement; dependency change; methodological expansion; methodological contraction; methodology supersession; methodology continuity; historical methodology preservation; historical version association; prediction-to-methodology-version linkage; methodology change traceability; methodology reactivation; methodology retirement; and complete version lineage.

Module Function

MVGM prevents organizations from treating Prediction Methodology as though it were a timeless and unchanging object.

Predictive systems evolve.

Models are retrained.

Rules change.

Weights are adjusted.

Experts enter or leave the process.

Data-processing stages change.

Dependencies are replaced.

AI systems are upgraded.

Aggregation mechanisms change.

A methodology that appears to retain the same name may therefore become materially different from the methodology that produced an earlier prediction.

MVGM establishes the governance layer required to distinguish:

same methodology, same version

from:

same methodology, changed version

and:

materially different methodology.

Methodology Version Object

Every governed methodology version should possess a sufficiently identifiable version object.

A Methodology Version Object may include:

methodology identifier; version identifier; effective state; predecessor version; successor version; change reference; change reason; materiality determination; affected methodological components; affected predictive logic; affected applicability conditions; activation state; retirement state; historical association.

The exact implementation may vary.

The governance requirement is that materially different methodology states remain distinguishable.

Initial Methodology Version

The first sufficiently governed form of a Prediction Methodology establishes an initial methodology version.

The initial version must be sufficiently linked to:

selected method; methodological structure; documented predictive logic; Method Applicability determination.

A methodology cannot possess meaningful version continuity where the original governed methodology state was never established.

Undefined Original Methodology $\neq$ Governable Version History

Version Identity

Each governed methodology version must remain uniquely distinguishable within the methodology lineage.

Version identity should prevent ambiguity between:

current version; historical version; experimental version; superseded version; retired version; reactivated version.

Version labels alone are not sufficient where they cannot be associated with the actual governed methodological state.

Version Number $\neq$ Version Governance

Methodology Change Identification

MVGM requires material changes affecting Prediction Methodology to become identifiable.

Potential changes may include:

selected method replacement; component addition; component removal; component substitution; sequence alteration; logic modification; threshold change; weighting change; aggregation change; model update; model retraining; human intervention change; dependency change; applicability condition change; methodological limitation change.

Not every change requires a new major version.

Every materially relevant change requires governance visibility.

Change Materiality

MVGM distinguishes changes according to whether they materially affect the methodology.

A change is materially relevant where it may alter:

how the methodology operates; what predictive outputs it can generate; how components interact; predictive logic; Method Applicability; interpretability of historical performance; comparability with earlier predictions.

Technical Change $\neq$ Material Methodological Change by Default

But:

Operationally Small Change $\neq$ Methodologically Immaterial Change by Default

Materiality depends on effect on the governed methodology, not merely implementation effort.

Minor Methodological Change

A change may remain within the same broader methodology while requiring version-level recognition.

Examples may include certain:

parameter adjustments; threshold modifications; configuration changes; bounded component updates; non-structural implementation changes.

A minor change must not be assumed merely because the organization labels it minor.

The materiality determination must be governed.

Major Methodological Change

A major methodological change may occur where modification materially alters the predictive method, structure, logic, applicability, or operational meaning of the methodology.

Potential examples include:

replacing one predictive model class with another; changing from single-model forecasting to an ensemble; introducing human override into an automated methodology; changing aggregation logic; replacing expert judgment with AI inference; adding simulation as a material predictive mechanism; changing prediction-market integration from supplementary to primary.

A major change may require a new methodology version or a newly identified methodology.

Model Replacement

Replacing a model inside Prediction Methodology may constitute a material methodology change.

This depends on the model's structural and predictive role.

A component replacement must not be treated as methodologically neutral merely because the surrounding system architecture remains unchanged.

Same Interface $\neq$ Same Prediction Methodology

Model Retraining

Retraining may materially change predictive behavior even where model architecture and name remain unchanged.

MVGM requires retraining to become version-governable where it can materially alter Prediction Methodology.

Potential considerations include:

training period; training population; training distribution; changed objectives; changed feature relationships; changed calibration behavior; altered predictive behavior.

Same Model Name $\neq$ Same Predictive Methodology State

Parameter and Threshold Change

A methodology may depend upon parameters or thresholds that materially influence prediction formation.

Where such values are changed, MVGM must determine whether the change:

preserves the current version,

requires a revised version,

or:

materially changes the methodology.

A numeric adjustment can be methodologically significant even where no source code changes.

Weighting Change

Changing weights among models, variables, experts, or predictive components may alter the meaning of the methodology.

Where weighting materially affects predictive output, changes must remain traceable.

Same Components + Different Weighting $\neq$ Same Methodological Behavior by Default

Aggregation Change

Changing how predictive signals are aggregated may create a materially different methodology.

Examples include changing from:

simple average $\rightarrow$ weighted average

majority vote $\rightarrow$ confidence weighting

human consensus $\rightarrow$ algorithmic aggregation

market signal $\rightarrow$ hybrid market-model aggregation

Aggregation change therefore requires explicit version governance.

Human Role Change

A Prediction Methodology may change materially where human participation changes.

Potential changes include:

addition of expert review; removal of human review; new override authority; changed adjustment rights; changed escalation role; changed final approval function.

Same Model + Different Human Authority $\neq$ Same Methodology by Default

Dependency Change

Replacing or materially changing a dependency may alter Prediction Methodology.

Dependencies may include:

external analytical services; computational tools; upstream models; simulation systems; external data-processing systems; market mechanisms; expert services.

MVGM governs the methodological consequences of dependency change.

The external dependency itself remains governed by its appropriate domain.

Method Applicability Change

A methodology version may remain structurally similar while its applicability changes.

For example:

the same methodology may become restricted to a narrower horizon; applicability may be withdrawn for one population; a new exclusion may emerge; a prior conditionality may become mandatory.

Where Method Applicability changes materially, the methodology version must preserve that change.

Same Method Structure $\neq$ Same Applicability State

Methodology Expansion

A methodology may expand through addition of new:

models; stages; signals; human roles; aggregation mechanisms; prediction-support mechanisms; domain coverage.

Expansion must not be assumed to preserve original methodology identity.

MVGM determines whether expansion creates:

same version,

new version,

or:

materially different methodology.

Methodology Contraction

Removing components or functionality may likewise alter methodology identity.

A simplified methodology may not remain equivalent to the methodology from which it was derived.

Reduced Methodology $\neq$ Original Methodology by Default

Version Transition

Every material transition between governed methodology versions should preserve a traceable relationship:

Version n → Material Change → Version n+1

where applicable.

A version transition should identify:

what changed; why it changed; when it changed;

what methodology elements were affected; whether applicability changed; which version superseded which.

Methodology Supersession

A new version may supersede an earlier methodology version.

Supersession means the earlier version is no longer the active methodology for new predictive use under the governed context.

It does not erase the historical version.

Supersession ≠ Deletion

A superseded methodology remains relevant to predictions produced under it.

Historical Methodology Preservation

Earlier methodology versions must remain preserved to the degree necessary to reconstruct which methodology governed historical predictions.

A current methodology description must not be retroactively applied to predictions generated under earlier versions.

Current Methodology ≠ Historical Methodology

This distinction is foundational to later predictive performance governance.

Prediction-to-Version Association

Each governed prediction should remain associable with the methodology version under which it was formed.

This relationship enables later governance to determine:

which method produced the prediction; which structure was active; which logic applied; which applicability state existed; which methodological limitations were known.

Prediction History Without Methodology Version History ≠ Complete Predictive History

Historical Performance Interpretation

A methodology may improve or degrade across versions.

Performance observed under one version must not automatically be attributed to another.

For example:

Version 1 accuracy ≠ Version 3 accuracy

unless sufficient evidence establishes comparable performance.

POFPS governs predictive performance.

MVGM ensures the correct methodology version remains available for that evaluation.

Version Continuity

Methodology Version Continuity requires preservation of relationships across:

Initial Version → Change → Revised Version → Supersession → Retirement / Reactivation

Continuity allows methodology evolution without erasing historical identity.

Parallel Methodology Versions

Multiple versions may exist simultaneously.

For example:

one version may remain active in one operational environment; another may be experimental; another may serve another population; another may remain approved for historical continuity.

MVGM permits parallel versions where their identity and applicability remain clear.

Latest Version ≠ Only Legitimate Version

Experimental Versions

Experimental or candidate methodology versions must remain distinguishable from approved operational versions where this distinction materially affects prediction governance.

Experimental use must not silently become operational methodology use without appropriate governance.

Methodology Retirement

A methodology version may be retired when it is no longer intended for active predictive use.

Retirement must preserve:

version identity; historical linkage; predecessor/successor relationships; predictions produced under the version; material methodological record.

Retirement ≠ Historical Erasure

Methodology Reactivation

A retired methodology version may potentially be reactivated where governance conditions permit.

Reactivation must not be assumed to restore prior applicability automatically.

A reactivated methodology may require:

applicability reassessment; dependency verification; limitation review; contextual review.

Historical Applicability ≠ Reactivated Applicability

Retrospective Methodology Drift

Organizations may unintentionally reinterpret an earlier prediction using the methodology currently in use.

MVGM prohibits retrospective methodology drift.

An earlier prediction must remain associated with the methodology that actually governed it at the time.

Later Methodological Improvement ≠ Permission to Rewrite Earlier Methodology

Version Traceability

A governed version lineage should allow reconstruction of:

what changed,

when it changed,

why it changed,

which version was active,

and:

which predictions were governed by that version.

Version traceability does not require unlimited technical archival.

It requires sufficient methodological reconstruction.

Methodology Version Sufficiency

Methodology version governance is sufficiently established when:

the active methodology version is identifiable; material changes are detectable; change materiality can be determined; predecessor and successor relationships are preserved; historical versions remain distinguishable; supersession does not erase history; Prediction-to-Version Association is possible; retired versions remain historically reconstructable; applicability changes remain traceable; material reactivation conditions can be reassessed.

Governance Requirements

MVGM requires that:

1. every governed Prediction Methodology possess identifiable version state;
2. initial methodology state be preserved;
3. material methodology changes remain detectable;
4. change materiality be determined rather than assumed;
5. model replacement and retraining be governed where methodologically material;
6. material logic, weighting, aggregation, dependency, or human-role changes remain traceable;
7. Method Applicability changes remain associated with the relevant version;
8. methodology expansion and contraction trigger version assessment where material;
9. supersession preserve historical methodology versions;
10. predictions remain associable with the methodology version that governed them;
11. retired methodology versions remain historically reconstructable;
12. reactivation not automatically inherit historical applicability;
13. retrospective methodology drift be prevented;
14. complete methodology version lineage remain traceable.

Governance Logic

MVGM applies the following governance logic:

Applicable Prediction Methodology

↓

Current Methodology Version Identified

↓

Methodological Change Detected?

If no:

Existing Version Preserved

If yes:

↓

Change Characterized

↓

Materiality Assessed

↓

Structure / Logic / Applicability / Dependency Impact Assessed

↓

Version Determination

↓

Existing Version Preserved / New Version Created / Methodology Reclassified

↓

Predecessor-Successor Relationship Recorded

↓

Historical Version Preserved

↓

Prediction-to-Version Association Enabled

↓

Governed Prediction Methodology

↓

PBAS — Prediction Basis Standard

Where version identity or lineage cannot be established:

Methodology Version Governance Gap → Clarification / Reconstruction / Reassessment

Minimum Implementation Framework

A minimum implementation of MVGM must establish:

A. Methodology Identifier

Identifier of the governed Prediction Methodology.

B. Active Version Identifier

The exact methodology version currently applicable.

C. Version State

At minimum:

Candidate → Active → Superseded → Retired → Reactivation Review → Reactivated

D. Predecessor Version

The methodology version from which the current version evolved, where applicable.

E. Change Record

Description of the material methodological change.

F. Change Category

Where applicable:

Structural / Logic / Applicability / Model / Parameter / Weighting / Aggregation / Human Role / Dependency / Other Material Change

G. Materiality Determination

At minimum:

Non-Material → Material Within Methodology → Major Methodological Change → Methodology
Reclassification Required

H. Effective Point

The point from which the methodology version became applicable.

I. Applicability State

The Method Applicability associated with the version.

J. Supersession Relationship

Whether and how the version supersedes another version.

K. Historical Preservation

Confirmation that prior versions remain reconstructable.

L. Prediction Association

Capability to associate a prediction with the methodology version governing it.

M. Reassessment Trigger

Conditions requiring new methodology version review.

Governance Outputs

MVGM may produce:

Methodology Version Identifier; active methodology version; methodology change record; materiality determination; predecessor-successor relationship; version lineage; supersession state; Method Applicability linkage; historical methodology record; retirement state; reactivation state; Prediction-to-Version Association; unresolved version-governance requirements.

The primary governed output is:

Governed Prediction Methodology Version

Completion of MVGM closes PMES and enables transition to:

PBAS — Prediction Basis Standard

Critical Distinctions

Version Number ≠ Version Governance

Undefined Original Methodology ≠ Governable Version History

Technical Change ≠ Material Methodological Change by Default

Operationally Small Change ≠ Methodologically Immaterial Change by Default

Same Interface ≠ Same Prediction Methodology

Same Model Name ≠ Same Predictive Methodology State

Same Components + Different Weighting ≠ Same Methodological Behavior by Default

Same Model + Different Human Authority ≠ Same Methodology by Default

Same Method Structure ≠ Same Applicability State

Reduced Methodology ≠ Original Methodology by Default

Methodology Update ≠ Historical Replacement

Supersession ≠ Deletion

Current Methodology ≠ Historical Methodology

Prediction History Without Methodology Version History ≠ Complete Predictive History

Latest Version ≠ Only Legitimate Version

Retirement ≠ Historical Erasure

Historical Applicability ≠ Reactivated Applicability

Later Methodological Improvement ≠ Permission to Rewrite Earlier Methodology

Enterprise Application

MVGM provides organizations with a governed methodology lineage across evolving predictive systems.

This becomes critical where production prediction systems change frequently through:

AI model upgrades; retraining cycles; new model providers; changed expert roles; new aggregation rules; updated thresholds; changed simulation components; redesigned analytical pipelines; altered applicability boundaries.

Without methodology version governance, an organization may maintain prediction records while losing the ability to establish which methodology actually produced them.

This makes later comparison, accountability, validation, auditing, and performance interpretation materially weaker.

MVGM preserves the methodological state behind each prediction.

Use Case 1 — AI Model Updated, Historical Predictions Re-Evaluated Under the Wrong Methodology

An organization uses an AI forecasting system for high-impact operational predictions.

During the year, the provider silently introduces several model updates and later replaces the underlying model entirely.

The user-facing product name remains unchanged.

Historical dashboards continue displaying predictions from before and after the updates as though they originated from one consistent methodology.

The new model appears significantly more accurate.

Management then concludes that the prediction system has demonstrated strong historical performance across the entire period.

The governance question is:

Can historical predictive performance legitimately be attributed to the current methodology when earlier predictions were produced by materially different model states that are no longer identifiable?

MVGM requires the methodology versions governing those predictions to remain distinguishable and associated with the predictions actually produced under them.

Without that association, the organization may incorrectly transfer performance from one methodology version to another.

Same Product Name ≠ Same Historical Prediction Methodology

Use Case 2 — Human Override Is Added Without Declaring a New Methodology Version

An organization originally operates an automated prediction model.

After several high-profile prediction failures, senior experts are given authority to manually adjust final predictions before publication.

No new methodology version is created because the underlying model itself has not changed.

Months later, some forecasts were produced directly by the model while others were materially altered by human judgment.

All are recorded under the same methodology version.

The governance question is:

Can predictions remain legitimately comparable under one methodology version when the authority to materially change predictive output was introduced without version recognition?

MVGM requires assessment of whether the new human role materially changes the Prediction Methodology.

If it does, the change must become visible in methodology version governance.

Same Model + New Predictive Authority $\neq$ Same Methodology by Default

Canonical Boundary

MVGM governs methodology version identity, methodology change detection, change materiality, version determination, supersession, historical preservation, Prediction-to-Version Association, retirement, reactivation, and methodology lineage. It does not govern predictive method selection, methodological structure definition, predictive logic documentation, initial Method Applicability determination, Prediction Basis, predictive assumptions, formation of a specific prediction, probability and uncertainty representation, feasibility, collective prediction legitimacy, Operational Reality, or predictive performance itself.

MVGM governs:

which methodology version governed the prediction and how that methodology changed over time.

It does not determine:

whether the prediction produced under that methodology was correct.

Canonical Closing Statement

A prediction cannot remain historically governable if the methodology that produced it disappears behind the methodology that replaced it.

MVGM preserves the methodological lineage necessary to know not only how a prediction was made, but exactly which version of that methodology made it.

11

NON-EXECUTABLE NOTICE

OOF® — OriginOpen® Foundation is a non-executable methodological authority.
This website and all publications are reference only.

CANONICAL LANGUAGE NOTICE

UCL™ is the canonical language layer of OOF® publications.
English is its carrier language.
In case of ambiguity, the published English HTML text prevails.

RIGHTS & PROTECTION



contact@originopenfoundation.org